PROJECT INFORMATION

Tittle	ESP32 VOICE CHAT
Version	v.0.1
Review by:	GH0XPWD
Notes:	Prototipo de ups con esp32

INDEX

1.- Portada
2.- Diagrama de bloques
3.-Circuito

Title		
Size A3	Number	Revision
Date: 7/27/2026		Sheet of
File: D:\PCB\IOT_micro_esp32s3\Portada.SchDoc		Drawn By:

Revisiones

V0.1

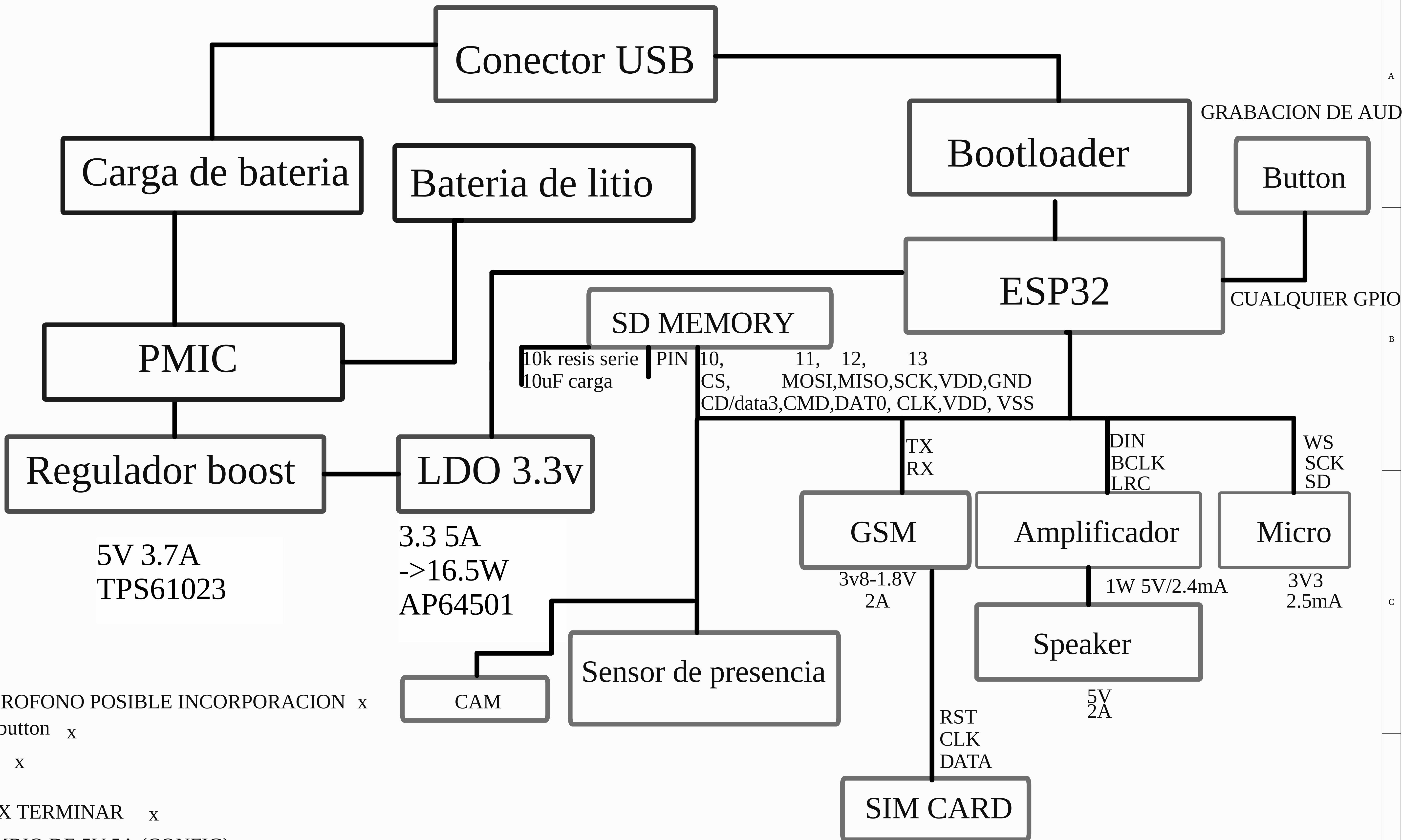
Mejorar footprints de resistencias a ML
Corregir polaridad de D+ D-
Corregir footprint de schocktty
Testear amplificador
Mejorar posicion de fuentes de regulacion para evitar emi
Mejorar serigrafia de la PCB

Usar 1.3mm de tamaño de serigrafia

V0.2

xiao cam s3 solo funciona en localhost
placa xiao tiene problema de temperatura

Title		
Size A3	Number	Revision
Date: 7/27/2026	Sheet of	
File: D:\PCB\..\Revisiones.SchDoc	Drawn By:	



MICROFONO POSIBLE INCORPORACION x
SW button x
SD x
SIM
MAX TERMINAR x
CAMBIO DE 5V 5A (CONFIG) x
GPIO presencia
GSM CONFIG UART AND ESD PROTECTION
CONECTOR SPEAKER x

Title			
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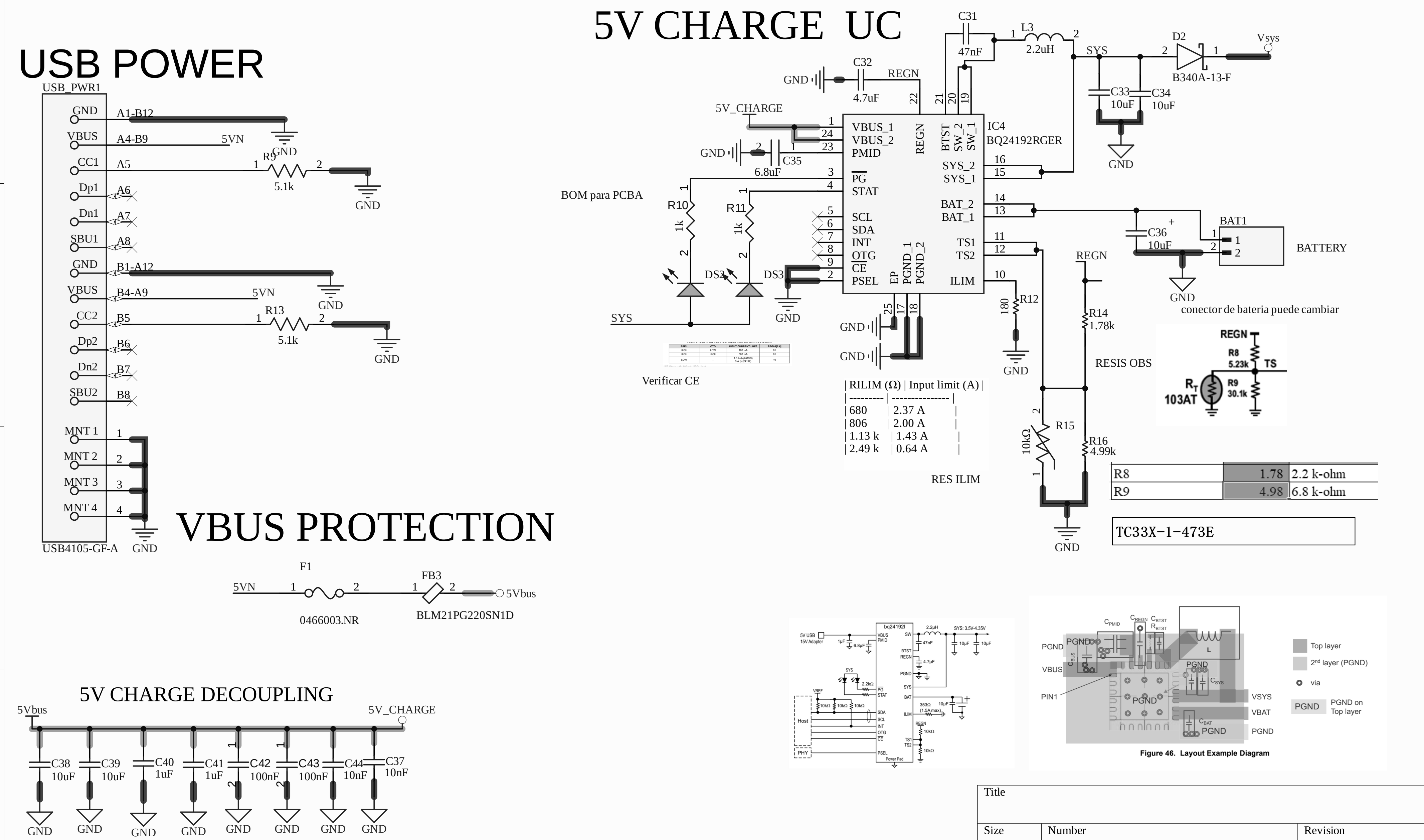
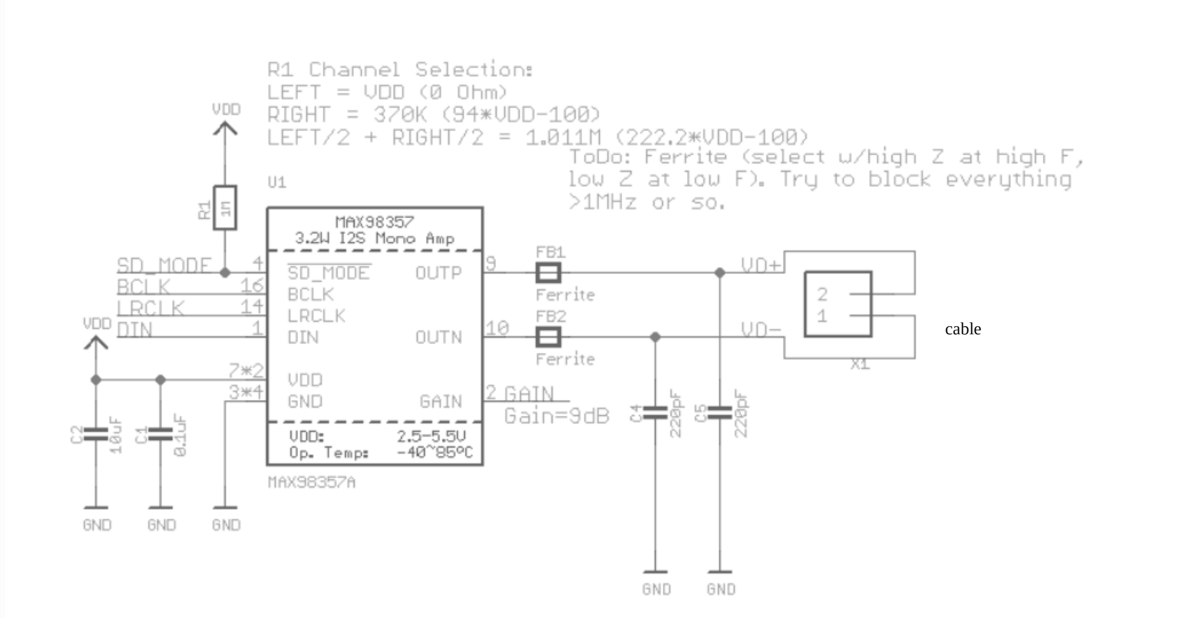


Figure 46. Layout Example Diagram

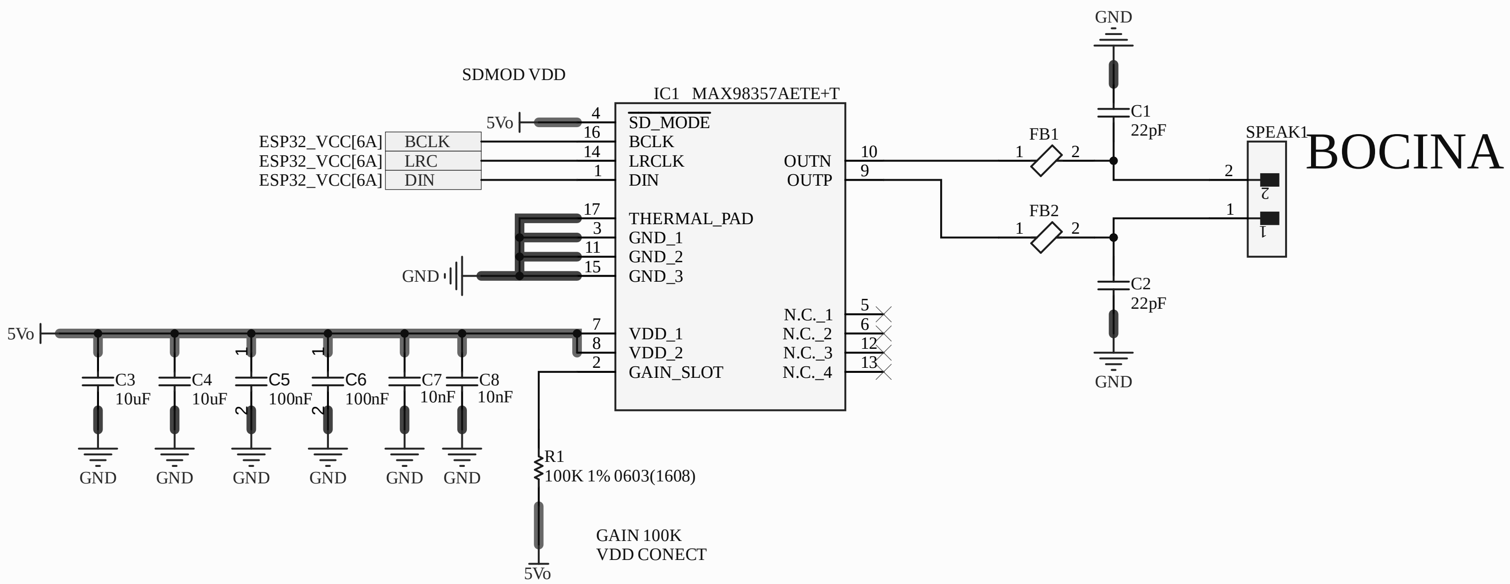
RILIM (Ω)	Input limit (A)
680	2.37 A
806	2.00 A
1.13 k	1.43 A
2.49 k	0.64 A

R8	1.78	2.2 k-ohm
R9	4.98	6.8 k-ohm

Title		
Size: A4	Number	Revision
Date: 7/27/2026		Sheet of
File: D:\PCB\..\PowerPath.SchDoc		Drawn By:



5V/2A



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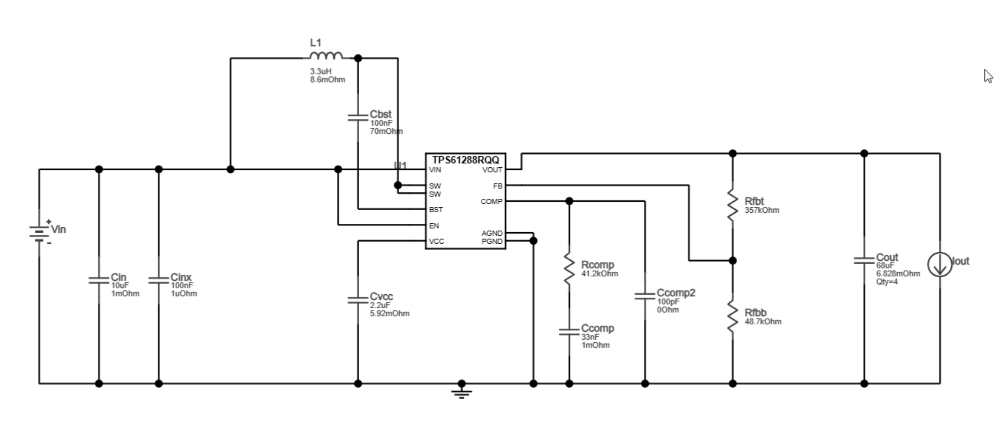
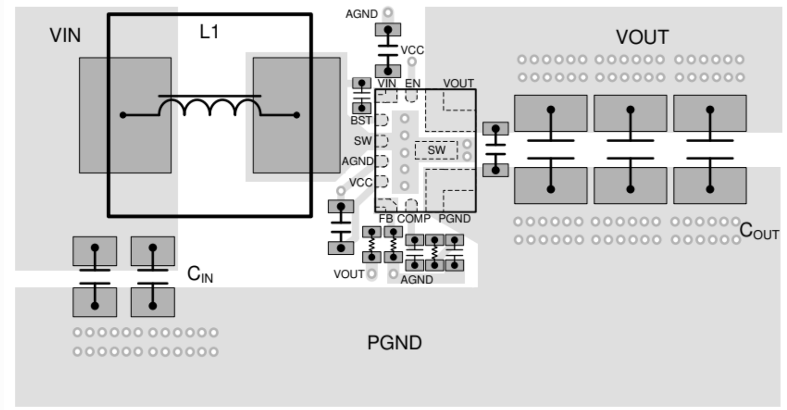
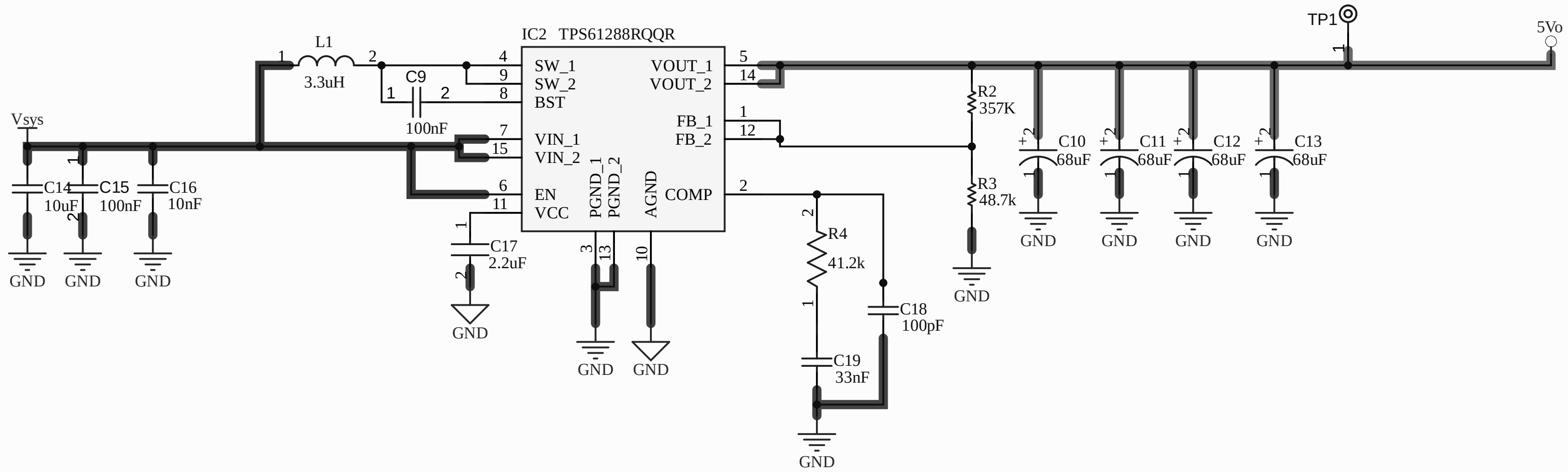
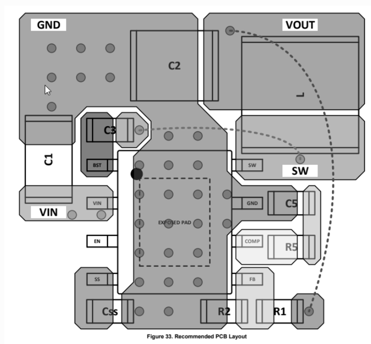


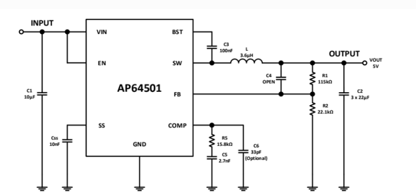


BOOST BAT/5V 1.5A





STEP DOWN to 3v3



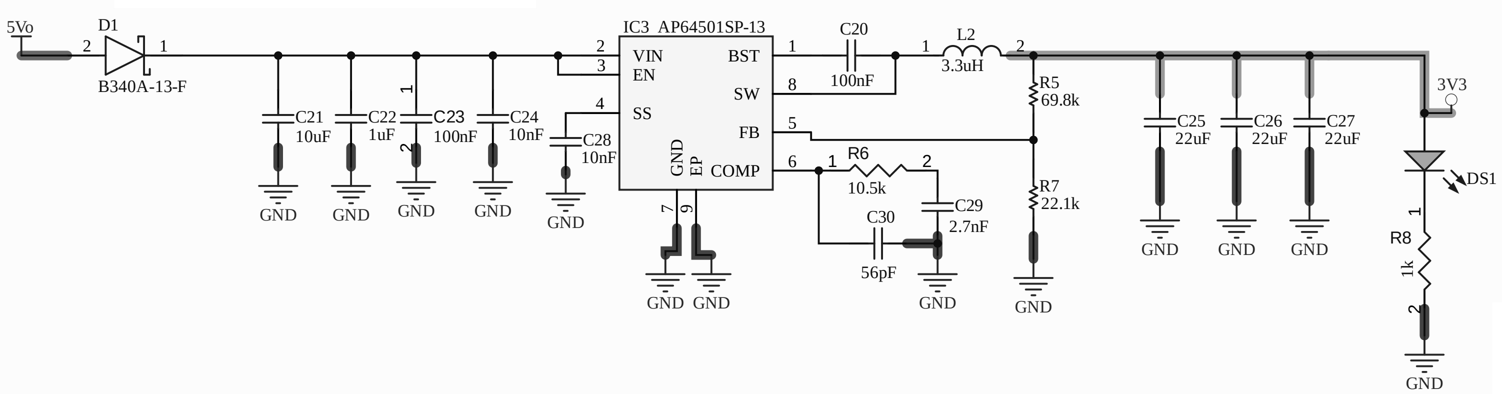


Table 1. Recommended Components Selections

AP64501										
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)	R5 (kΩ)	C5 (nF)	C6 (pF) (Optional)
1.2	11.0	22.1	1.5	10	3 x 22	100	OPEN	3.74	2.7	150
1.5	19.6	22.1	2.2	10	3 x 22	100	OPEN	4.75	2.7	120
1.8	27.4	22.1	2.2	10	3 x 22	100	OPEN	5.62	2.7	100
2.5	47.5	22.1	3.3	10	3 x 22	100	OPEN	7.87	2.7	68
3.3	69.8	22.1	3.3	10	3 x 22	100	OPEN	10.50	2.7	56
5.0	115.8	22.1	3.6	10	3 x 22	100	OPEN	15.80	2.7	33
12.0	309.0	22.1	10.0	10	3 x 22	100	OPEN	37.40	2.7	15

Title		
Size: A3	Number	Revision
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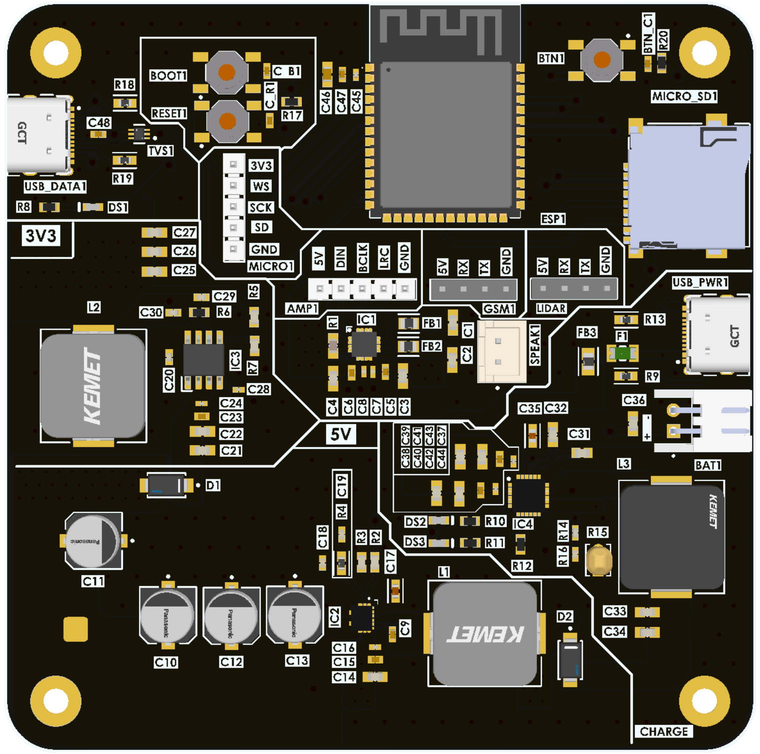
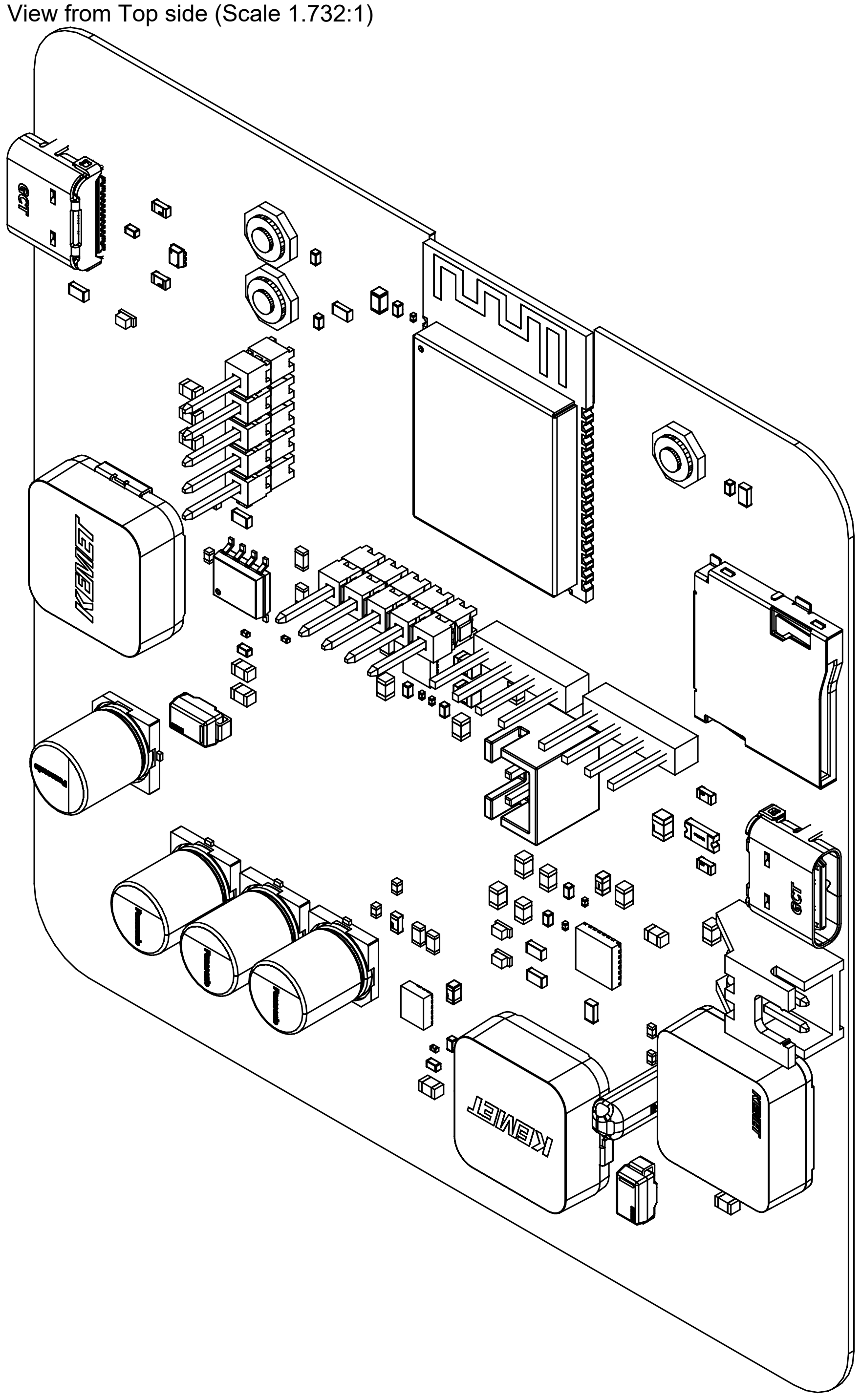
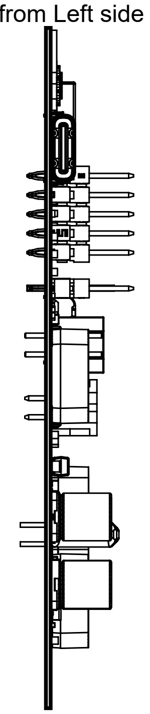
Realistic View



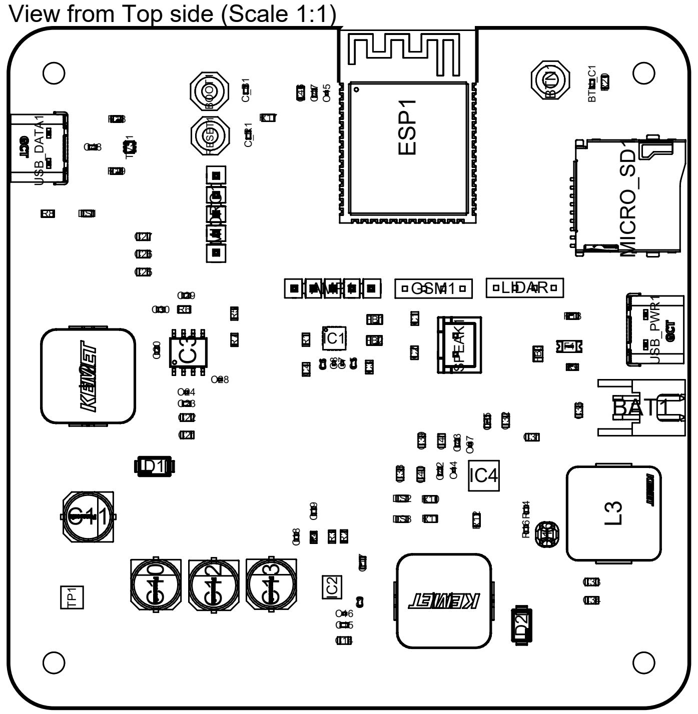
View from Top side (Scale 1.732:1)



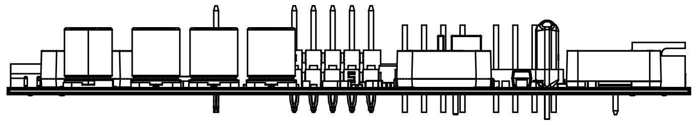
View from Left side (Scale 1:1)



View from Top side (Scale 1:1)



View from Front side (Scale 1:1)



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NEXT ASSY	USED ON
APPLICATION	

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DIMENSIONS ARE IN INCHES
TOLERANCES:
FRACTIONAL±
ANGULAR: MACH± BEND ±
TWO PLACE DECIMAL ±
THREE PLACE DECIMAL ±

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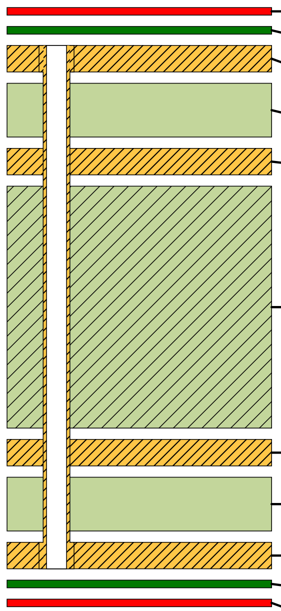
SCALE: 1:1	WEIGHT:	SHEET 1 OF 3

1

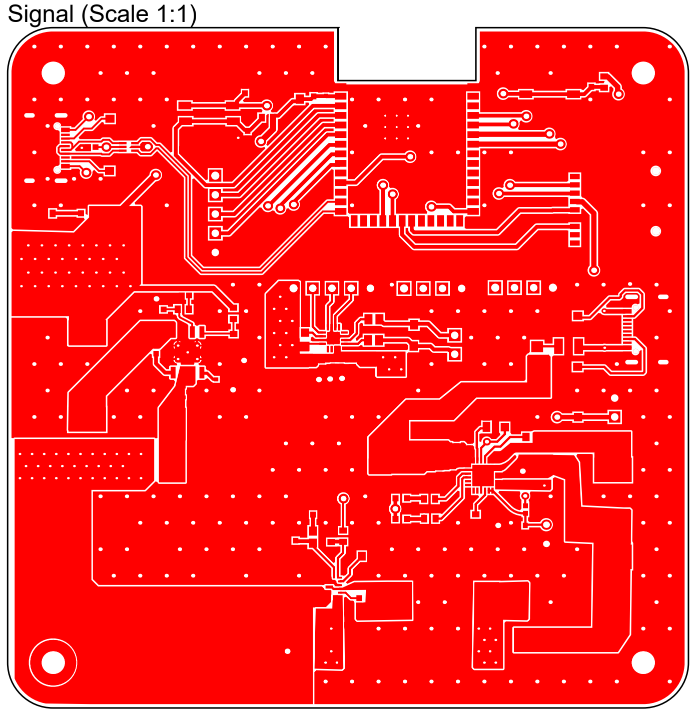
Layer Stack Legend



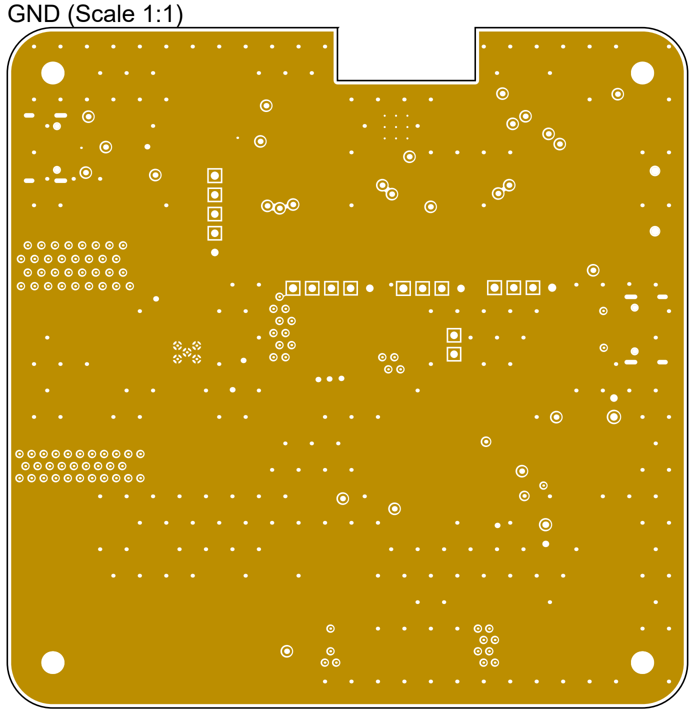
Layer	Thickness	Type
Top Overlay		Legend
Top Solder	0.01mm	Solder Mask
Signal	**0.04mm**	**Signal**
	0.07mm	*Dielectric*
GND	**0.04mm**	**Signal**
	0.32mm	*Dielectric*
PWR	**0.04mm**	**Signal**
	0.07mm	*Dielectric*
GND/SIGNAL	**0.04mm**	**Signal**
Bottom Solder	0.01mm	Solder Mask
Bottom Overlay		Legend

Total thickness: 0.62mm

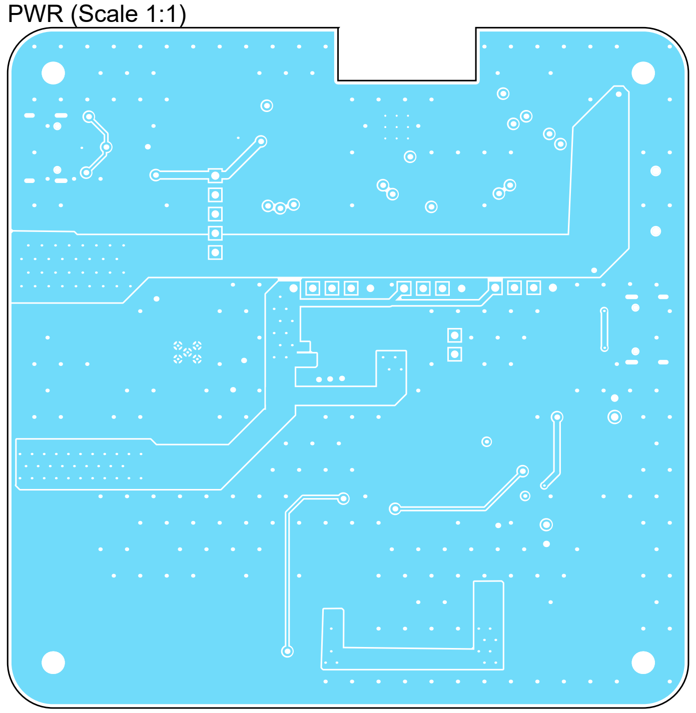
Signal (Scale 1:1)



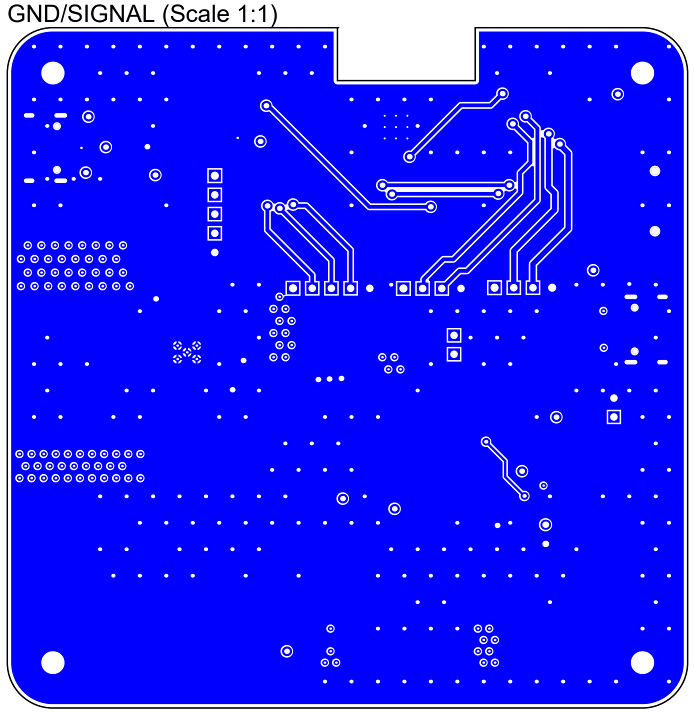
GND (Scale 1:1)



PWR (Scale 1:1)



GND/SIGNAL (Scale 1:1)



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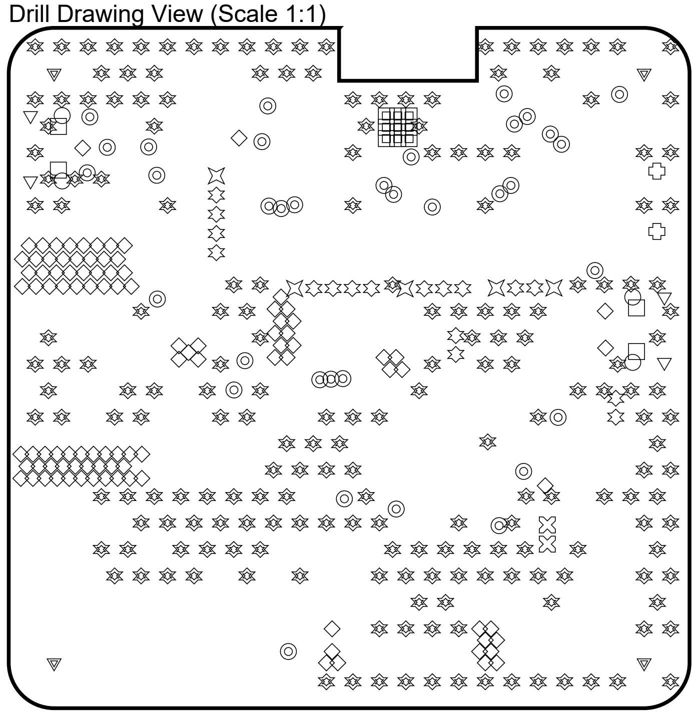
SIZE	DWG. NO.	
SCALE: 1:1	WEIGHT:	SHEET 2 OF 3

Drill Drawing View (Scale 1:1)



Drill Table

Symbol	Count	Hole Size	Plated	Hole Tolerance
▣	9	0mm	Plated	
◇	101	0mm	Plated	
	205	0mm	Plated	
□	4	1mm	Non-Plated	
◎	35	1mm	Plated	
	2	1mm	Plated	
	2	1mm	Non-Plated	
	17	1mm	Plated	
	5	1mm	Plated	
▽	4	1mm	Plated	
○	4	2mm	Plated	
	4	3mm	Plated	
	392 Total			

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MFG APPR.		
Q.A.		

COMMENTS:

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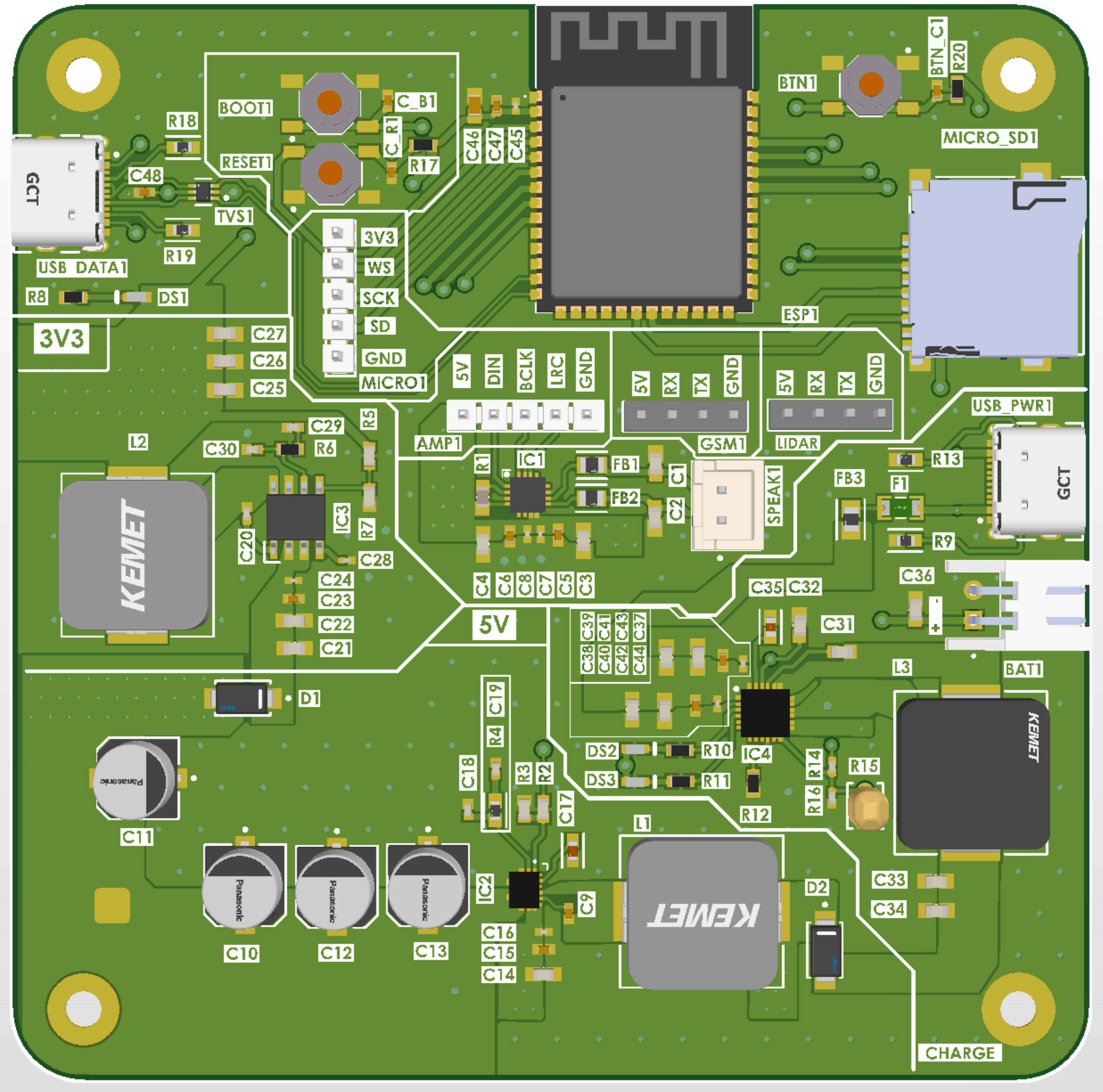
SIZE | DWG. NO.

SCALE: 1:1 | WEIGHT: | SHEET 3 OF 3


BOOT1
RESET1
C_B1
C_R1
R17
R18
C48
TVS1
R19
GCT
USB_DATA1
R8
DS1
3V3
C27
C26
C25
3V3
WS
SCK
SD
GND
MICRO1
C46
C47
C45
ESP1
BTN1
BTN_C1
R20
MICRO_SD1
5V
DIN
BCLK
LRC
GND
AMP1
5V
RX
TX
GND
GSM1
5V
RX
TX
GND
LIDAR
USB_PWR1
R13
FB3
F1
GCT
R9
L2
C29
R5
C30
R6
IC3
R7
C20
C28
C24
C23
C22
C21
KEMET
R1
IC1
FB1
FB2
C1
C2
SPEAK1
C4
C6
C8
C7
C5
C3
5V
C35
C32
C36
C31
C38
C39
C40
C41
C42
C43
C44
C37
L3
BAT1
KEMET
D1
C19
R4
C18
R3
R2
C17
DS2
R10
DS3
R11
IC4
R14
R16
R15
R12
C11
C10
C12
C13
IC2
C9
C16
C15
C14
L1
KEMET
D2
C33
C34
CHARGE